MIL-P-21397C(SHIPS)
28 November 1971
SUPERSEDING
MIL-P-21397B(SHIPS)
23 August 1960
(See 6.5)

MILITARY SPECIFICATION

PROPORTIONING UNIT, CHEMICAL (FOR DISTILLING PLANTS NAVAL SHIPBOARD USE)

1. SCOPE

1.1 This specification covers a chemical proportioning unit consisting of a motor driven, positive displacement reciprocating type pump with attached liquid supply tank for use with shipbosxd distilling plants.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-185 - Bearings, Roller, Cylindrical; and Bearings, Roller, Self-Aligning,
FF-B-187 - Bearing, Roller, Tapered.
TT-G-490 - Cleaning Methods end Pretreatment of Ferrous Surfaces for Organic Coatings.
TT-P-645 - primer, Paint, Zinc-Chromate, Alkyd Type.
WW T-799 - Tube, Copper, Seamless, (For Use with Solder-Type Or Flared-Tube Fittings).
GGG-P-781 - Puller, Mechanical Puller Attachment, Mechanical, and Puller Set, Mechanical.

MILITARY

MIL-P-735 - Primer Coatings, Exterior Metal (Yellow), Zinc Chromate, Alkyd Type.
MIL-D-1OOO - Drawings, Engineering and Associated List.
MIL-D-l000/2 - Drawings, Engineering and Associated List.
MIL-F-1183 - Fittings, Tube Cast Bronze, Silver-Brazing.
MIL-C-2212 - Controllers, Electric Motor, A.C., or D.C. and Associated Switching Devices, Naval Shipboard.
MIL-G-2695 - Glasses, Gage, Round and Flat, Reflex, Under 125 Pounds Pressure.
MIL-D-5480 - Data, Engineering and Technical: Reproduction Requirements For.
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification,
MIL-M-15071 Manuals, Technical: Equipments and Systems Content Requirements for.
MIL-E-15090 Enamel, Equipment, Light-Gray (Formula No, 11).
MIL-P-15137 Provisioning Technical Documentation For Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-C-15328 Primer (Wash) Pretreatment, Blue (Formula No. 117 for Metals).
MIL-P-16789 Preservation, Packaging, Packing and Marking of Pumps General, and Associated Repair Parts.
MIL-M-17059 Motors, 60-Cycle, Alternating-Current, fractional H.P. (Shipboard Use).
MIL-P-17545 Primer Coating, Alkyd-Red Lead Type Fomula No. 116 and Formula No. l16D.
MIL-P-17549 Plastic Laminates, Fibrous Glass Reinforced Marine Structural.
MIL-M-17556 Motor, Direct-Current, Fractional HP (Shipboard Use).
MIL-C-18480 Coating Compound Bituminous, Solvent, Coal Tar Base,
MIL-I-45208 Inspection System Requirements.

STANDARDS

MILITARY

MIL-STD-100 Engineering Drawing Practices.
MIL-STD-278 Fabrication Welding and Inspection; end Casting Inspection and Repair For Machinery, Piping and Pressure Vessels in Ships-of the United States Navy,

DRAWINGS

MILITARY

B-214 - Root Connections for Attaching Piping.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer,)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein, Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A276 - Hot Rolled and Cold-Finished Stainless and Heat Resisting Steel Bars.
A283 - Low and Intermediate Tensile Strength Carbon-Steel plates of Structural Quality.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa, 19103.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

SOCIETY OF AUTOMOTIVE ENGINEERS
SAE-1117 - Hot Rolled Carbon Steel Bars

(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, New York l0017.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies,)

3. REQUIREMENTS

3.1 General design. The design and construction of the pump and reducer shell be the lightest and most compact possible consistent with accessibility for maintenance and the replacement of wear parts. The pump shall be designed for a discharge pressure of 400 pounds per square Inch gage (p.s.i.g.) even though the operating pressure will be lower (see 4.3.4), The pump shall have a maximum capacity of 2.5 gallons per hour (gph) at 90 psig discharge pressure, adjustable within 5 percent from 0.5 gph to 2.5 gph. The pump shell be preset at the factory at 0.5 gph for a simplex pump and 1.0 gph for a duplex pump unless otherwise specified (see 6.1).

3.2 Materiel.

3.2,1 Material, general, Materiel shall be as specified herein. Material not definitely specified shell be of the best quality used for the purpose in commercial practice, and shall be subject to review and approval by the command or agency concerned. Material shall be free from all defects and imperfections that might adversely affect the serviceability of the finished product.

3.2.2 Substitution of material.. The use of materiels other than those indicated will be considered when the contractor shows the necessity for the substitution and the material proposed is readily obtainable. Substitution of materials will require the specific approval of the command or agency concerned in each case.

3.2,3 Material for pumps shall be in accordance with table I.

Table I - Material for pumps.

Application	Material.	Applicable document
Base plate	Steel, low carbon	ASTM A283
Casting	Close grained, cast iron 40,000 lbs. tensile	---
Cylinder	Cres, class 316	ASTM A276
Plunger	Cres, class 4400 hardened to C58-60R	ASTM A276
Valve body	Ores, class 316	ASTM A276

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Table I - Material for pumps. (cont,d.)

Application	Materiel	Applicable document
Valve seat	Cres, class 316	ASTM A276
Check valve ball	Cres, class 329	---
Vent valve ball	Cres, class 440C	ASTM A276
Worm	Steel (carburized to Rockwell 52C.)	SAE 1117
Worm wheel (gear)	Bronze (84.0 ~~Cu~~ 10.0 Sn-2.5-3.5Ni)	---

3,3 General construction.

3.3.1 Unless otherwise specified (see 6.1), pumps shall be mounted below the supply tank and the entire assembly shall be capable of passing through an opening 24 inches by 30 inches. The proportioning unit base shall be a separate piece with provisions for bolting to a mounting platform forming part of the supporting structure of the supply tank.

3.3.2 In no case shall a piece of equipment be supported fron more than one plane.

3.3.3 Bolts subject to shear stress shall be installed in holes with a minimum of clearance to eliminate impacting and subsequent bolt failure.

3.3.4 Baseplates shall be sufficiently rigid to permit handling and installation of unit onboard ship without disturbibg alignment of assembled units and such that normal distortion weaving or vibration of supporting structures onboard ship cannot cause misalignment between pump and driver.

3.3.5 Each component part of the proportioning unit shall be supported directly by the baseplate. Pump couplings shall be flexible end shall be supplied on all procurements,

3.3.6 All bearing and seating surfaces of castings, brackets or pump feet shall be finish machined.

3.3.7 Bearings.

3.3,7.1 Radial bearings. Radial bearings may be of the cup and cone, roller or sleeve type.

3.3.7.2 Roller bearings for radial loads and installation thereof shall be in accordance with FF-B-185 and FF-B-187, as applicable.

3.3.7.3 Crankpin and wrist pin bearings shell be of self-aligning construction. Sleeve bearings shall be of the fixed or self-aligning type. Ample bearing surface shall be provided for each bearing.

3.3.8 Bearing lubrication. Externally mounted bearings shall be lubricated in accordance with FF-B-185 or FF-B-187, as applicable.

3.3.9 Service operation. The proportioning unit, in addition to performing satisfactorily in the normal operating position, shall be designed and assembled to operate under the most unfavorable combination of the following conditions:

(a) Inclined angles of 30 degrees,
(b) Inclined angles up to 45 degrees for a minimum of 3 minutes,
(c) 60 degree roll angle when the maximum roll period is 12 seconds.

NOTE : The the to complete one roll cycle is from one extreme to the other and return.

3.4 Liquid end of pump shell be readily removable from pump assembly to provide for ease of replacement of liquid end components. Pump cylinder shell be provided with a manually operated ball type vent valve for purging system in the event it becomes air or gas bound, Pump plunger shall be self-aligning and supported by a guide bushing and not by the plunger packing. Plunger stroke adjustment shall preferably be of the slotted crank arm type with lead screw adjustment. The scale shall be of a percentage calibrated type. Lost motion stroke adjustment shall not be used and the plunger speed shell not exceed 60 strokes per minute.

3.5 Supply tank.

3.5.1 The tank shall be cylindrical, 18 inches in diameter and 32 inches high. It shall have an indicated capacity of 24 gallons and a 2-1/2 gallon reserve capacity below the zero point. A drain connection shall be provided in the tank bottom, furnished with a valve for connecting to a 3/4 inch drain hose. The pump suction shall be 2 inches above the tank bottom,

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3.5.2 The tank shall. be of polyester fiberglass with a minimum thickness of 3/16 inches of grade W or better in accordance with MIL-P-17549.

3.5.3 The tank shall be provided with a cover which shall be bolted and gasketed to flanges on the top of the tank. The cover shall be provided with an opening for filling the tank. This opening shell be not less than 3 inches in diameter with a 2 inch lip extending into the tank and shall be provided with a vented tank cover, suitably secured,

3.5..4 A gage glass is not required. A suitable scale shall be integral with the tank graduated from O to 24 gallons in 1 gallon increments to indicate the amount of solution in the tank,

3.5.5 The foundation shall be fabricated to provide firm support for the tank with piping and fittings located beneath the tank.

3.5.6 Piping and valves. Discharge line shall be flanged in accordance with Drawing B-214,

3.5.7 Piping shell be 1/2 inch copper tubing in accordance with WW-T-799 with fittings in accordance- with MIL-F-1183. Suction piping shall be unrestricted except for shutoff valves and strainer.

345.7.1 A vacuum drag bypass connection shall be installed in the piping system, with valves, to provide emergency supply of feed treatment solution to the evaporator in case the pump unit is inoperable, A needle valve shall be provided for close regulation of the flow through the bypass. Valves shall be provided for isolation of the pump from the tank and bypass. The feed treatment solution shall flow through the strainer to either the pump suction or the bypass connection.

3.5.8 Needle valves shall be long stemmed to insure accurate metering adjustment. Commercial bronze valves may be used.

3.5.9 Relief valves.

3.5.9.1 Relief valves shall be standard commercial 150 pounds per square inch (p.s.i.) maximum working pressure, nonferrous material with bronze trim.

3.5.9.2 Relief valves shall be fitted on the discharge side with setting at 50 p.s.i, There shall be no intervening valves between relief valve and relief valve outlet.

3.6 Threaded parts. Threaded parts shall conform to Handbook H28.

3.7 Welding shall be in accordance with MIL-STD-278.

3.8 Painting.

3.6.1 Pumps. All external unmachined surfaces of ungalvanized or noncadmium plated ferrous metal parts shell be thoroughly cleaned and coated with one coat of pretreatment conforming to MIL-C-15328, and one coat of zinc-chromate primer, conforming to TT-P-645 or one coat of red lead primer conforming to MIL-P-17545, followed by a finish coat of light gray equipment enamel conforming to MIL-E-15090.

3.8.2 Tank foundation. All surfaces shell be prepared and painted in accordance with 3.8.1.

3.8.3 Painting of external surfaces of any non-ferrous parts will not be required,

3.9 Motor.

3.9.1 The motor shall be as specified (see 6.1) and in accordance with service C of MIL-M-17059 for fractional horsepower (hp.) alternating current (a.c,) motors and service C of MIL-M-17556 for fractional hp. direct current (d.c.) motors.

3.9.1.1 Motor rating. The hp. rating of each motor shall be not less than 1/4 hp.

3.9.1.2 Controllers. Motor controllers and pressure switches shell be in accordance with MIL-C-2212 and shall have the characteristics as specified (see 6.1).

3.9.2 Reduction gears. Gears shall be worm type with material in accordance with 3.2.3. The gears shall have a maximum backlash of 0.010 inch as indicated at a 3 inch radius on the low speed shaft with the input shaft locked.

3.9.2.1 Reduction gear and housing shall be integral with the pump frame and connected thereto by connecting rod bolts.

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3.10 Identification plates.

3.10.1 Identification plates shall be furnished on each pump. These plates shall be type A or B in accordance with MIL-P-15024 and MIL-P-15024/5, except that identification plates of aluminum or plastic shall not be ueed.

3.10.2 Identification plates shall be firmly secured to the equipment with corrosion-resistant metallic screws or drive nails.

3.10.3 Pump identification plates shall contain the following data:

(a) Manufacturer's name.
(b) Manufacturers model or type and size.
(c) Servioe application.
(d) Manufacturer's serial number.
(e) Salient design characteristics:
    (1) Capecity in gph.
    (2) Discharge pressure.
    (3) Brake hP.
    (4) Test pressure.
(f) Contract number
(g) Federal stock number.
(h) Section for Defense Contract Administration Services (DCAS) stamp.
(i)
(j) Component identification number (CID No.) (9 digits).

3.11 Drawings.

3.11.1 Drawings shall conform to categories E, G and H, form 2 of MIL-D-1000 and types II and III of MIL-D-1000/2.

3.11 .1.1 Assembly drawing. An assembly drawing shall be furnished covering complete proportioning units including pump, driver, and accessories. This drawing shall include complete performance data and performance curves, and also a table of weights of components and complete unit.

3.11.1.2 Sectional assembly drawing. A sectional assembly drawing of pump shall be furnished with complete list of material, All running clearances shall be shown. The bill of material shall include an indication of each part to be furnished as an onboard repair part.

3,11.1.3 Envelope drawing. The requirements of this drawing may be combined with the assembly drawing.

3.11.2 The certification data shall be in accordance with MIL-D-1000/2. In addition, a list of onboard repair parts furnished under the contract or order shall also be provided in the certification data. If required, the certification data may be combined with the assembly drawing.

3.11.3 Drawings of motors and motor controllers shall contain all the information required on drawings by the applicable motor and controller specifications.

3.11.4 Drawings and certification data sheets for approval shall be blueprints or equal in accordance with type II, class 1 of MIL-D-5480.

3.11.5 Final drawings and certification data sheets (tracings) shall be in accordance with MIL-D-1000/2.

3.11.6 The center of gravity of the complete unit shall be indicated on the assembly drawing.

3.12 Teohnical manuals.

3,12.1 Technical manuals shall be in accordance with type I of MIL-M-15071.

3,12.1.1 Operating instructions will be enclosed in clear waterproof plastic liner and supplied with unit.

3,12.2 Identification piece numbers of parts referred to in technical manuals shall be identical to the identification piece numbers shown on approved. master drawings.

3.12.3 Technical manuals shall contain reproductions of outline drawings of all units.

3.12.4 Preliminary technial manuals submitted for approval to the command or agency concerned shall include all proposed sections of the manuals.

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3.12,5 Two copies of' technical manufals shall be packed with each proportioning tit. Additional copies shall be furnished as specified (see 6.1). If final technical manuals are not available at the time units are shipped, then two copies of preliminary technical manuals shall be packed with each proportioning unit. The final manuals shall be delivered at a later time as specified in MIL-M-15071.

3.13 Repair parts and tools.

3.13.1 Onboard. Repair parts shall be furnished and processed in accordance with method B (paragraph 3.3.2) of MIL-P-15137 (see 6.3) and as specified in the contract.

3.13.2 Each set or repair parts and tools specified in the contract or order shall contain the parts listed in table II as a minimum. Unless otherwise specified in the contract or order, the quantity of repair parts per set shall be based on one pump per ship.

Table II - Onboard repair parts.

Name or part	Quantity per pump
Connecting rod, complete with bearings	1
Wrist pin	1
Plunger:	
For simplex pump	1
For duplex pump	2
Check valve, discharge and suction, complete consisting of two check valve balls, one check valve body, upper, and one check valve body, lower:	
For simplex pump	1
For duplex pump	2
Bushing:	
For simplex pump	2
For duplex pump	4
Packing set (requirement to include shelf life expectancy):	
For simplex pump	1
For duplex pump	2
Bearings, reducer:	
High speed	1
Low speed	1

3.13.2.1 On all procurements the manufacturers part number (vendor's number w/FSCM's) shall be entered in addition to the prime contractor part number and manufacturer's Federal Code (FSCM).

3.13.3 Repair parts for motors, and controllers, shall be furnished in accordance with the applicable component specifications or as specified in the contract or order.

3.73.4 Special tools and wrenches.

3.13.4.1 Each set of onboard repair parts shall include a complete set of all special tools and wrenches required for maintenance of the pumps, gears and drivers. Special tools are defined as those tools not listed in the Federal Supply Catalog, (Copies of this catalog may be consulted in the office of the DCAS).

3.13.4.2 One set of special operating tools and wrenches shall be furnished with each proportioning unit, and shall include all wrenohes and tools required for the operation of the complete proportioning units, Each set of special operating tools and wrenches shall be furnished in separate boxes. If there are no more than two such tools per set they may be attached to the pumps in a secure but readily accessible manner, rather than boxed .

3.13.4.3 Each tool or wrench shall be indelibly marked indicating the intended purpose.

3.13.4.4 Tools which are available in the Federal Supply Catalog, such as common wrenches and standard pullers, will not be required, Standard complete or limited sets of pullers or parts of puller sets in accordance with GGG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed, and included in the list of material.

3.13.4,5 Each box containing onboard repair parts shall contain a list entitled "List of onboard repair parts and tools", The list shall be in a format for use on outline drawings and in the technical manuals. It shall be not less than nominal 8-1/2 by 11 inches in size.

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3.13.4.5,1 The format of the Iist shall include a heading and columns of date for the items listed. The heading shall include titles and applicable entries as follows:

Reproduced from drawing number_____
Number of ships_____
Application_____
Contract number_____
Manufacturer_____
Quantities are for_____units per ship
The columns shall include:
Piece number_____
Name of part_____
Quantity_____
Drawing number_____
Federal stock number_____
Additional columns may be used as applicable.

3.13.4.5.2 The list shall be so treated as to be resistant to oil, water and fading.

3.14 Stock. The selection, stock numbering and quantities or repair parts for stock shall be determined and processed accordance with MIL-P-15137 (see 6.3).

3,14,1 Unless otherwise specified in the contract or order, the cognizant supply demand control point for repair parts for stock shall be Ships Parts Control Center, Mechanicsburg, Pennsylvania.

4. QUALITY ASSURANCE PROVISIONS

4,1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed neceesary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The inspection system which the supplier is required to maintain as provided in the inspection clause of the contract or order shall be in accordance with MIL-I-45208 (see 6.1 and 6.3).

* 4.2 Classification of tests. Tests shall be classified as follows:

(a) Performance acceptance (see 4.3).
(b) Hydrostatic (see 4.4).

4.3 Performance acceptance test,

4.3.1 One pump with motor, supply tank and all appurtenances, connected and in place as intended for shipboard operation, of each class, design or size on each contract or order shall be operated at maximum speed and capacity, with the pumped fluid at maximum pressure for 45 minutes taking three equally spaced bearing temperature readings to ascertain that bearing temperatures are normal. The pump shall be operated at minimum capacity, one-half capacity and three-quarter capacity until the test values being measured stabilize, also at maximum pressure.

4.3.2 Sufficient data shall be taken during the tests specified in 4.3.1 to show complete operating characteristics of pump and driver, Capacity shall be expreseed in gph, discharge in psig and speed in strokes per minute.

4.3.3 Government shall conduct performance tests at random of 20 percent of the total proportioning units on the contract, and others to be manufacturer's standard test,

4.3.4 The capacity of the pump may be determined by any convenient method that will insure obtaining results with probable errors not greater than 2 percent. Capacity of the proportioning unit shall be determined at 100 percent, 75 percent, 50 percent and 25 percent of rating at 90 p.s.i.g. All gages shall, be calibrated before and after the tests or more often, if necessary. Relief valve setting shall be checked (see 3.5,9.2). During test, pumps shall be stalled by high pressure at least four times (see 4.4.1).

4.3.5 Reports,

4.3.5.1 Complete performance acceptance test reports, including copies or recorded test data covering the following, shall be furnished as specified in 4.3.5.2:

(a) Test report of driving unit,
(b) Test report of pump.

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(c) Test report on the combined assembled unit including a definite report on the suitability of the driving unit for a specific application and harmony of assembly with all independent units going to make up the complete equipment.

4.3.5.2 The report specified in 4.3.5.1 shall be distributed as follows:

(a) Three to NAVSEC (to be forwarded via the DCAS).
(b) One for DCAS district.
(c) One for branch DCAS office (if any).
(d) Two to each supervisor of shipbuilding concerned (one for supervisor's files, the other for the shipbuilder).
(e) One to the Commander of each ship building yard concerned, for Navy Yard built ships, when specified (see 6.1),

4.4 Hydrostatic tests.

4.4.1 Pump cylinders and valve chambers shall be tested hydrostatically to the maximum design pressure.

4.4.2 The tank shell be filled with water to the top to ascertain that the tank is water tight.

4.5 Inspection of Preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direot Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4.)

5.1 Cleaning, Reservation. packaging packing and marking. The equipment, accessories, repair parts, tools and technioal manuals shall be prepared for the level of shipment as specified (see 6.1) in accordance with MIL-P-16789. Contact preservatives shall not be applied to the liquid supply tank, gage glass and nonferrous items.

5.1.1 Drawings. Drawings shall be prepared for the level of shipment as specified (see 6.1) in accordance with MIL-D-1000/2.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Presetting of pump (see 3.1).
(c) Where pump shall be mounted if other than below supply tank (see 3.3.1).
(d) Voltage of motor type of enclosure and current characteristics for motor and controller (see 3.9.1 and 3.9.4).
(e) Additional number of technical manuals required (see 3.12.5).
(f) Quality assurance requirements (see 4.1.1).
(g) Number of copies of report to be furnished (see 4.3.5.2).
(h) Level of shipment required (see 5.1.1 and 5.1.2).

6.2 Proportioning unit alone may be purchased in cases where an existing tank can be used as a supply tank.

6.3 Management control system documents. The following management control system documents, should be included on DD form 1660:

(a) MIL-P-15137 (see 3.13.1 and 3.14).
(b) MIL-I-45208 (see 4.1.1).

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

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6.5 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOYS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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